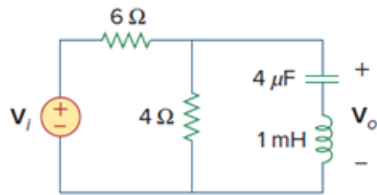


Problem

Find the bandwidth and center frequency of the band-stop filter of Fig. 14.89.

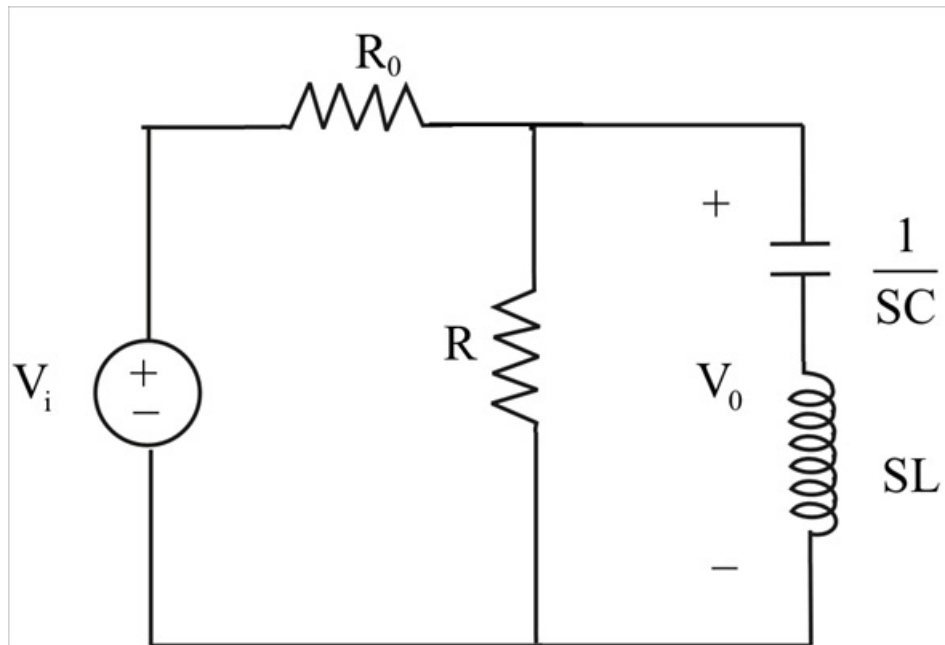
Figure 14.89:



Step-by-step solution

Step 1 of 7

Consider the circuit below



Step 2 of 7

$$Z(s) = R \parallel \left(sL + \frac{1}{sC} \right)$$

$$Z(s) = \frac{R \left(sL + \frac{1}{sC} \right)}{R + sL + \frac{1}{sC}}$$

$$Z(s) = \frac{R(1 + s^2 LC)}{1 + sRC + s^2 LC}$$

Step 3 of 7

$$H(\omega) = \frac{V_o}{V_i} = \frac{Z(s)}{Z(s) + R_o}$$

$$H(\omega) = \frac{R(1+s^2LC)}{R_o + sRR_oC + s^2LCR_o + R + s^2LCR} \dots\dots\dots (1)$$

Step 4 of 7

$$Z_{in} = R_o + Z(s)$$

$$Z_{in} = R_o + \frac{R(1+s^2LC)}{1+sRC+s^2LC}$$

Put $s = j\omega$

$$Z_{in} = \frac{R_o + j\omega RR_oC - \omega^2LCR_o + R - \omega^2LCR}{1 - \omega^2LC + j\omega RC}$$

$$Z_{in} = \frac{(R_o + R - \omega^2LCR_o - \omega^2LCR + j\omega RR_oC)(1 - \omega^2LC - j\omega RC)}{(1 - \omega^2LC)^2 + (\omega RC)^2}$$

$$I_m(Z_{in}) = 0,$$

$$-\omega RC [R_o + R - \omega^2LCR_o - \omega^2LCR] + \omega RR_oC(1 - \omega^2LC) = 0$$

$$-\omega RC [(R_o + R - \omega^2LCR_o - \omega^2LCR) - R_o + \omega^2R_oLC] = 0$$

$$R_o + R - \omega^2LCR_o - \omega^2LCR - R_o + \omega^2R_oLC = 0$$

$$\omega^2LCR = R$$

$$\omega^2LC = 1$$

$$\omega^2 = \frac{1}{LC}$$

$$\omega = \frac{1}{\sqrt{LC}} \dots\dots\dots (2)$$

Step 5 of 7

L, C values substitute in Equation (2) we get,

$$\omega = \frac{1}{\sqrt{(1 \times 10^{-3})(4 \times 10^{-6})}}$$

$$\omega = 15.811 \text{ k rad/sec}$$

From equation – (1)

$$H(\omega) = \frac{R(1 - \omega^2LC)}{R_o + j\omega RR_oC - \omega^2LCR_o + R - \omega^2LCR}$$

$$H_{\max} = H(0) = \frac{R}{R + R_o} \text{ and}$$

$$H_{\max} = H(\infty) = \frac{R}{R + R_o}$$

Step 6 of 7

At ω_1 and ω_2 , $|H(\omega)| = \frac{1}{\sqrt{2}} H_{\max}$

$$\frac{1}{\sqrt{2}} \left(\frac{R}{R + R_0} \right) = \left| \frac{R(1 - \omega^2 LC)}{R_0 + j\omega R R_0 C - \omega^2 L C R_0 + R - \omega^2 L C R} \right|$$

$$\frac{1}{\sqrt{2}} = \left| \frac{(R + R_0)(1 - \omega^2 LC)}{R_0 + j\omega R R_0 C - \omega^2 L C R_0 + R - \omega^2 L C R} \right|$$

$$\frac{1}{\sqrt{2}} = \frac{(R + R_0)(1 - \omega^2 LC)}{\sqrt{(R_0 + R - \omega^2 LC(R_0 + R))^2 + (\omega R R_0 C)^2}}$$

R, R_0, L & C values substitute in above equation we get,

$$\frac{1}{\sqrt{2}} = \frac{(4 + 6)(1 - \omega^2(1 \times 10^{-3}) \times (4 \times 10^{-6}))}{\sqrt{(6 + 4 - \omega^2(1 \times 10^{-3}) \times (4 \times 10^{-6}))(6 + 4))^2 + (\omega \times 4 \times 6 \times (4 \times 10^{-6}))^2}}$$

$$\frac{1}{\sqrt{2}} = \frac{10 - \omega^2(4 \times 10^{-8})}{\sqrt{(10 - \omega^2 \times (4 \times 10^{-8}))^2 + (\omega(96 \times 10^{-6}))^2}}$$

$$\frac{1}{2} = \frac{(10 - \omega^2(4 \times 10^{-8}))^2}{(10 - \omega^2 \times (4 \times 10^{-8}))^2 + (\omega(96 \times 10^{-6}))^2}$$

$$\frac{1}{2} = \frac{100 + \omega^4(16 \times 10^{-16}) - \omega^2 80 \times 10^{-8}}{(100 + \omega^4 \times (16 \times 10^{-16}) - \omega^2(80 \times 10^{-8})) + \omega^2(9216 \times 10^{-12})}$$

$$\frac{1}{2} = \frac{100 + \omega^4(16 \times 10^{-16}) - \omega^2 80 \times 10^{-8}}{100 + \omega^4 \times (16 \times 10^{-16}) - \omega^2 7.91 \times 10^{-7}}$$

$$100 + \omega^4 \times (16 \times 10^{-16}) - \omega^2 7.91 \times 10^{-7} = 200 + \omega^4(32 \times 10^{-16}) - \omega^2 160 \times 10^{-8}$$

$$\omega^4(16 \times 10^{-16}) - \omega^2 8.09 \times 10^{-7} + 100 = 0$$

$$\omega^2 = 2.9 \times 10^8 \text{ \&}$$

$$\omega^2 = 2.15 \times 10^8$$

Therefore,

$$\omega_1 = 14.663 \text{ k rad/sec \&}$$

$$\omega_2 = 17.029 \text{ k rad/sec}$$

Step 7 of 7

Therefore,

$$\text{The band width } (B) = \omega_2 - \omega_1$$

$$\text{The band width } (B) = (17.029 \times 10^3) - (14.663 \times 10^3)$$

$$\text{The band width } (B) = 2.399 \text{ k rad/sec}$$